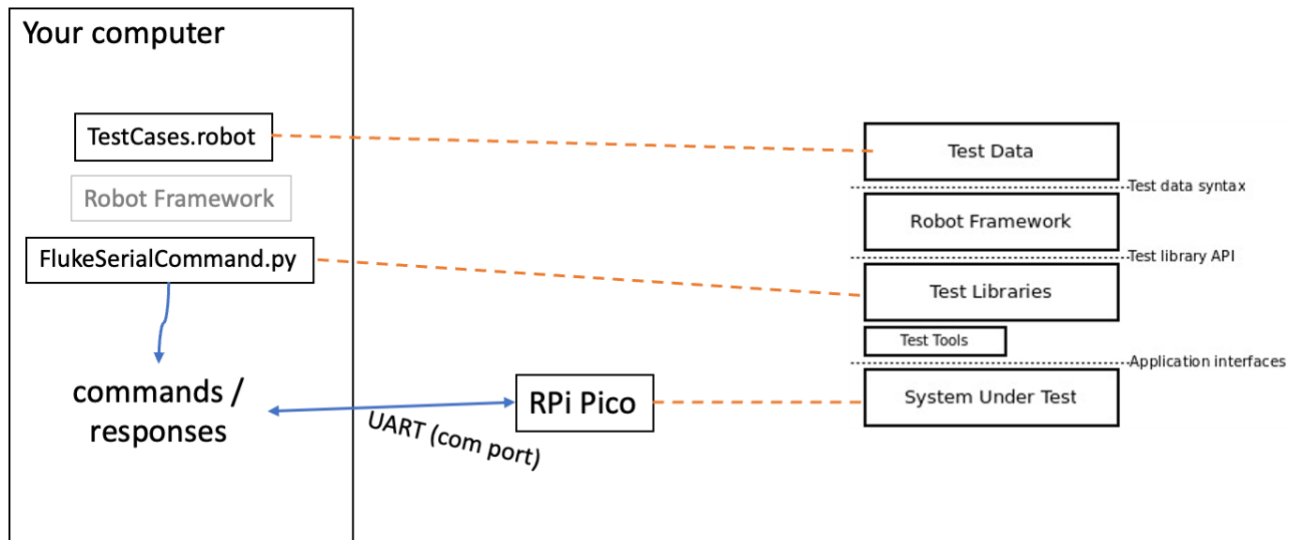


Using your own library

In this exercise you get a python library as a starting point for interfacing a device using USB serial port. In this assignment you need to make some changes to the library and implement Robot Framework tests that interface Fluke Prosim 8 patient monitor tester, which is simulated by RPi Pico.



Program the RPi Pico

Hold down BOOTSEL, the only button, on Pico when plugging it to a USB port to put Pico in programming mode. When Pico is in programming mode it shows up as a mass storage device (removable drive) on your computer. Programming takes place simply by copying **prosim_simulator.uf2** to the removable drive. After programming is finished Pico reboots automatically and starts running code from the flash. This makes the removable drive disappear and depending on the operating system a warning about unsafe removal may be displayed. The warning can be safely ignored.

Try out the simulator

Open the serial port with Serial monitor or a terminal program of your choice. Study **prosim8_-_communications_interface_ver3.17.pdf**, the user manual of the patient monitor tester. Try out different commands and see how the simulator responds before writing any Robot Framework tests. Note that the simulator does not echo the received characters back. To see what you typed you need to switch local echo on in your terminal program.

Install pyserial

To be able to use the provided test library (**FlukeSerialCommand.py**) you need to install **pyserial** package:

```
pip install pyserial
```

Edit the library

Edit the comport name in **FlukeSerialCommand.py** to match the port in your system.

Implement tests and improve library

Implement two test cases:

- Connect To Fluke
- Set Simulator State

Connect To Fluke tests that it is connected to Fluke Prosim by reading the identity of the device. The test case must check that response contains “PROSIM8” and output the firmware version with Log -keyword. After that the test case reads the serial number and writes it to log. Note that when a device starts up some garbage characters can be received during startup so your first command response may be an error even though the command you send is correct. You can simulate this by entering some junk chars with terminal before starting the test. Remember to close terminal before test.

Set Simulator State switches to remote control mode, sets temperature to 36.5 degrees, sets the heart rate (adult) to 78bpm, sets standard deviation to 0.05, and turn on ECG wave. Test case must validate all responses from the simulator.

Improve library by making the serial port a parameter. Set the port that you usually use in a variable that is passed as a parameter to the library. You can also make any other changes you need to complete the assignment.

Deliverables

Show to the instructor:

- Test case file
- Improved library
- Command line to run the test with overridden serial port. Set your default to non-existing port before the demo
- That your tests run without failing

Common issues with pyserial

If you get an error about non-existing keywords, it means that the FlukeSerialCommand failed to initialize properly. That can happen if pyserial package is not installed or if the comport name don't match the port in your system.

On windows there is an issue with serial port permissions. It seems that serial ports don't get closed quickly enough after Suite setup which causes “permission denied” errors.

To fix that, you can set the library scope to 'SUITE' in the library. Setting scope to 'GLOBAL' also works.

```
import serial
@library
class FlukeSerialCommand():
    ''' Library for interacting with Fluke Prosim 8
    ROBOT_LIBRARY_SCOPE = 'SUITE'
```

Note for Mac users: Mac assigns port names based on the location of the device in the USB topology. If you always connect the Pico to the same port (in the same USB hub) then device is always assigned the same name.